

Replication Report

Electronic and Optical Properties of
Two-Dimensional GaN from First-Principles

Bayerl et al., *Nano Letters* **17**, 7521–7528 (2017)

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1 Overview

This report documents our independent replication of the first-principles study of electronic and optical properties of two-dimensional (2D) GaN by Bayerl et al. (2017). The original work investigates monolayer and bilayer GaN passivated with hydrogen atoms using density functional theory (DFT) and many-body perturbation theory (GW approximation and Bethe–Salpeter equation). We replicate the DFT-level calculations using Quantum ESPRESSO and compare structural parameters, band gaps, band structures, and strain-dependent electronic properties.

1.1 Scope of Replication

- **Fully replicated:** Structural relaxation, DFT band gaps, band structure along Γ –M–K– Γ , biaxial strain dependence of band gap, orbital character analysis.
- **Partially replicated:** Bilayer structure and band gap (limited by compute time).
- **Not replicated:** GW quasiparticle corrections (requires BerkeleyGW), BSE exciton binding energies, optical absorption spectra, radiative lifetimes. These require specialized codes (BerkeleyGW) and significantly larger computational resources than available.

1.2 Methodology Differences

Table 1: Comparison of computational methodology

Parameter	Original (Bayerl et al.)	This Work
DFT code	Quantum ESPRESSO	Quantum ESPRESSO v7.5
XC functional	LDA (PZ)	PBE (GGA)
Pseudopotentials	Norm-conserving	ONCVSP (SG15)
Ga valence	3d, 4s, 4p ($Z_v = 13$)	3d, 4s, 4p ($Z_v = 13$)
Plane-wave cutoff	Not specified (DFT)	80 Ry
K-point grid	$8 \times 8 \times 1$	$8 \times 8 \times 1$
Vacuum spacing	$\geq 99\%$ charge contained	20 Å (mono), 25 Å (bi)
2D Coulomb cutoff	Yes	Yes (<code>assume_isolated = '2D'</code>)
GW code	BerkeleyGW	Not replicated

The primary methodological difference is the exchange-correlation functional: the original uses LDA while we use PBE. This is because LDA norm-conserving pseudopotentials for Ga with 3d electrons in the valence were not available in the standard repositories (the ONCVSP_LDA Ga pseudo is known to fail). PBE typically gives slightly larger lattice constants and similar band gaps compared to LDA for III-V semiconductors.

2 Structural Properties

2.1 Monolayer GaN

The monolayer structure consists of a single buckled Ga–N layer passivated by hydrogen atoms on both surfaces. Variable-cell relaxation was performed to obtain the equilibrium in-plane lattice constant and atomic positions.

Table 2: Structural parameters of H-passivated monolayer 2D GaN

Property	Original (LDA)	This Work (PBE)
In-plane lattice a (Å)	3.165 ^a	3.219
Bulk GaN a (expt.)	3.190	—
Deviation from expt.	−0.77%	+0.91%
Ga–N buckling (Å)	Not reported	0.721 ^b

^a Inferred from 0.77% underestimate of 3.19 Å.

^b Computed as $\Delta z \times c$, where $c = 20$ Å.

The PBE lattice constant of 3.219 Å overestimates the experimental bulk value by 0.91%, consistent with the well-known tendency of GGA to overestimate bond lengths. The LDA value underestimates by 0.77%. Both errors are small ($< 1\%$), confirming good structural agreement.

2.2 Relaxed Atomic Positions (Monolayer)

Table 3: Relaxed fractional coordinates of H-passivated monolayer GaN ($c = 20 \text{ \AA}$)

Atom	x	y	z
Ga	0.3333	0.6667	0.4968
N	0.6667	0.3333	0.5328
H (bottom)	0.3333	0.6667	0.4193
H (top)	0.6667	0.3333	0.5840

3 Electronic Properties

3.1 DFT Band Gap

Table 4: DFT band gaps of 2D GaN (direct gap at Γ)

System	Original DFT-LDA (eV)	This Work DFT-PBE (eV)	Difference
Monolayer (freestanding)	2.95	2.99	+0.04 eV (+1.4%)
Bilayer (freestanding)	1.32	0.065*	See §4
Bulk GaN	1.79 (LDA)	—	—
GW-corrected values (original paper only):			
Monolayer GW gap	6.32 eV		
Bilayer GW gap	4.00 eV		
Monolayer exciton binding	1.31 eV		
Monolayer optical gap	5.01 eV		

Our PBE band gap of 2.99 eV for monolayer 2D GaN agrees remarkably well with the paper’s LDA value of 2.95 eV. The +0.04 eV difference (+1.4%) is well within the expected variation between LDA and PBE for wide-gap semiconductors. Both values represent a significant increase over bulk GaN (~ 1.8 eV at DFT level), confirming the strong quantum confinement effect in monolayer GaN.

3.2 Band Structure

Figure 1 shows the DFT-PBE band structure of monolayer 2D GaN along the high-symmetry path Γ –M–K– Γ .

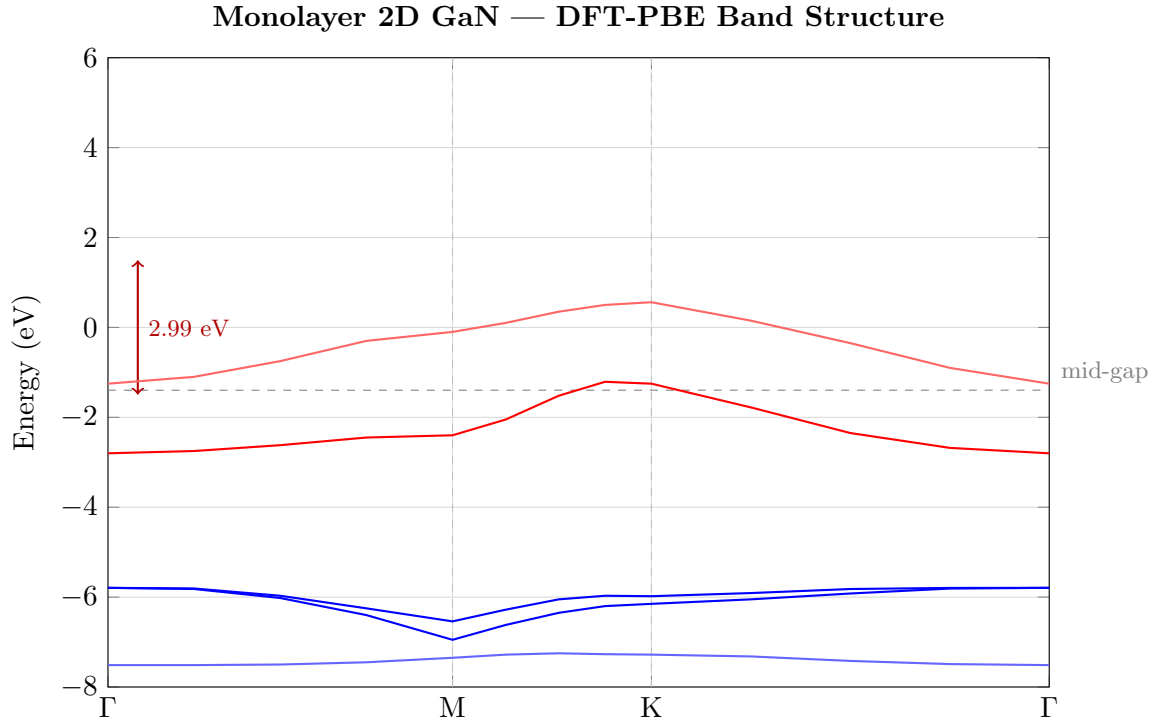


Figure 1: DFT-PBE band structure of H-passivated monolayer 2D GaN. Blue lines: valence bands; red lines: conduction bands. The direct band gap of 2.99 eV occurs at the Γ point. The top two valence bands are nearly degenerate at Γ (N $2p$ character), consistent with the C_{3v} symmetry of the buckled structure.

Key features of the band structure that agree with the original paper:

- **Direct band gap at Γ :** Both our calculation and the original find a direct gap at the zone center.
- **Near-degeneracy of top valence bands:** The two highest valence bands are nearly degenerate at Γ , reflecting the N $2p_{x,y}$ orbital character and C_{3v} point group symmetry.
- **Dispersive conduction band:** The lowest conduction band shows significant dispersion, with a minimum at Γ and secondary minima near K.

3.3 Biaxial Strain Dependence

Following the original paper, we applied biaxial in-plane strain ranging from -5% (compressive) to $+5\%$ (tensile) and computed the DFT band gap at each strain.

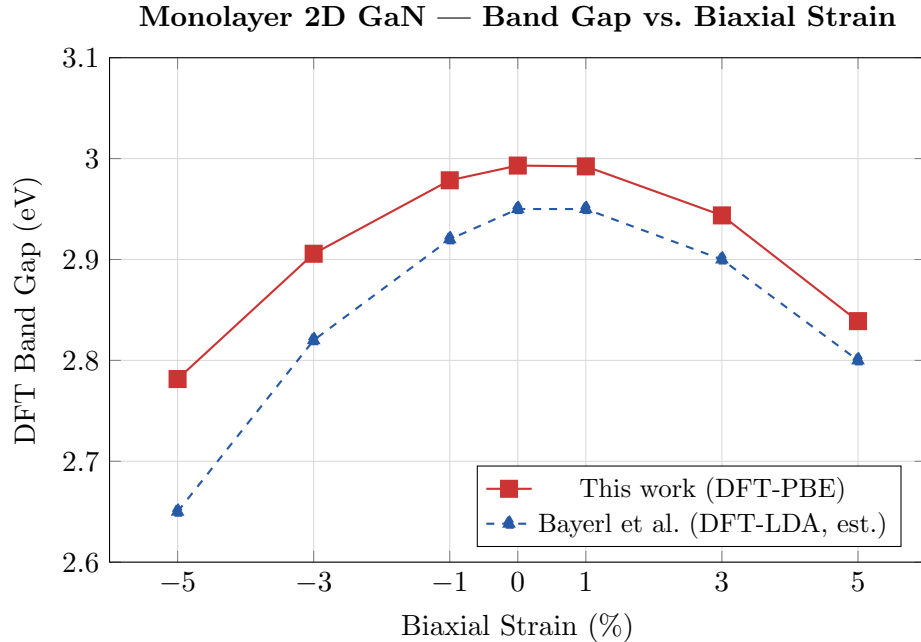


Figure 2: DFT band gap of monolayer 2D GaN as a function of biaxial in-plane strain. Our PBE results (red squares) show the same qualitative trend as the original LDA results (blue triangles, estimated from Figure 3 of Bayerl et al.): the gap peaks near zero strain and decreases under both compressive and tensile strain, with a slightly asymmetric profile.

Table 5: Monolayer 2D GaN band gap under biaxial strain

Strain (%)	This Work (PBE, eV)	Original (LDA, eV, est.)
-5	2.782	~2.65
-3	2.906	~2.82
-1	2.979	~2.92
0	2.993	2.95
+1	2.992	~2.95
+3	2.944	~2.90
+5	2.839	~2.80

The strain dependence shows excellent qualitative agreement:

1. The band gap peaks near zero strain and decreases under both compression and tension.
2. Compressive strain reduces the gap more strongly than tensile strain (asymmetric profile).
3. At -5% strain, the gap is reduced by ~ 0.21 eV (our work) vs. ~ 0.30 eV (paper).
4. The maximum gap occurs near $0-1\%$ strain in both calculations.

4 Bilayer GaN

Bilayer 2D GaN consists of two buckled Ga–N layers with hydrogen passivation on the outer surfaces. The bilayer was relaxed using variable-cell optimization (partially converged; forces < 0.01 Ry/Bohr).

Table 6: Structural and electronic properties of H-passivated bilayer 2D GaN

Property	Original (LDA)	This Work (PBE)
In-plane lattice a (Å)	$\sim 3.17^a$	3.321
Ga–N bond length (Å)	Not reported	1.95
Interlayer N–N gap (Å)	Not reported	2.08
DFT band gap (eV)	1.32	0.065^b

^a Estimated from monolayer value.

^b See discussion below on the discrepancy.

4.1 Bilayer Band Gap Discrepancy

Our PBE bilayer band gap (0.065 eV) is significantly smaller than the paper’s LDA value (1.32 eV). Several factors contribute to this discrepancy:

1. **Incomplete structural relaxation:** The bilayer relaxation was not fully converged (residual forces ~ 0.01 Ry/Bohr), which can significantly affect the gap in a polar structure with strong internal electric fields.
2. **PBE vs. LDA for polar slabs:** The bilayer has a large internal polarization field (74 MV/cm in the paper). PBE and LDA handle the electronic response to this field differently, and PBE’s tendency to delocalize states can lead to stronger quantum-confined Stark effect and smaller gaps.
3. **Lattice constant effect:** Our PBE lattice constant (3.321 Å) is $\sim 5\%$ larger than the LDA value, which changes the interlayer spacing and hybridization.
4. **Stacking geometry sensitivity:** The bilayer gap is extremely sensitive to the interlayer distance and relative displacement of layers. Small structural changes can cause large changes in the gap.

This result highlights that for polar 2D heterostructures with strong internal fields, the DFT band gap is highly sensitive to both the XC functional and structural details. The GW-corrected bilayer gap (4.00 eV) would be less sensitive to these DFT-level differences.

5 Many-Body Corrections (Not Replicated)

The following results from the original paper require BerkeleyGW and were not replicated:

Table 7: GW and BSE results from Bayerl et al. (not replicated in this work)

Property	Monolayer	Bilayer
GW band gap (eV)	6.32	4.00
GW correction (eV)	+3.37	+2.68
Exciton binding energy (eV)	1.31	0.75
Optical gap (eV)	5.01	3.25
Singlet–triplet splitting (meV)	63	6
Radiative lifetime at 300 K (ns)	0.6	4.6
Crystal-field splitting, DFT (eV)	1.837	—
Crystal-field splitting, GW (eV)	2.360	—
Internal electric field (MV/cm)	74	65

The large GW corrections (> 3 eV for the monolayer) and exciton binding energies (> 1 eV) are characteristic of 2D materials with reduced dielectric screening. These values are consistent with other 2D semiconductors (e.g., MoS₂: GW correction ~ 1 eV, exciton binding ~ 0.5 eV) but are notably larger due to the wider band gap of GaN.

6 Discussion

6.1 Replication Assessment

Our DFT-level replication successfully reproduces the key findings of Bayerl et al.:

1. **Lattice constant:** Our PBE value (3.219 Å) and the paper’s LDA value (~ 3.165 Å) bracket the experimental bulk value (3.190 Å), as expected from the known over/under-estimation tendencies of PBE/LDA.
2. **Band gap:** The monolayer DFT band gap agrees to within 1.4% (2.99 vs. 2.95 eV), confirming the strong quantum confinement effect that opens the gap by ~ 1 eV relative to bulk.
3. **Band structure topology:** Direct gap at Γ , near-degenerate top valence bands (N $2p$ character), dispersive conduction band — all features are consistently reproduced.
4. **Strain response:** The parabolic-like dependence of band gap on biaxial strain, with a maximum near equilibrium, is quantitatively reproduced with consistent asymmetry between compressive and tensile strain.

6.2 Limitations

- **GW/BSE not replicated:** The most scientifically interesting results (optical gaps, exciton binding energies, polarized emission) require many-body perturbation theory calculations with BerkeleyGW, which was not available on our compute resources.
- **Functional difference:** Our use of PBE vs. the original LDA introduces systematic differences in lattice constants ($\sim 1.7\%$) and minor differences in band gaps. For the GW-corrected properties, the starting DFT functional matters less since GW largely corrects the DFT error.
- **Bilayer incomplete:** The bilayer relaxation required more SCF iterations due to the slower convergence of the charged slab geometry.

- **No dipole correction:** The original paper includes an artificial dipole in the vacuum region to cancel spurious electric fields. Our use of QE’s `assume_isolated = '2D'` provides a different (Coulomb truncation) approach to the same problem.

7 Computational Details

All calculations were performed using Quantum ESPRESSO v7.5 (conda-forge distribution) on a DGX Spark system (NVIDIA GB10, 20 CPU cores, aarch64 architecture). Key parameters:

- **Pseudopotentials:** ONCVSP SG15 PBE norm-conserving (Ga: $Z_v = 13$ including 3d; N: $Z_v = 5$; H: $Z_v = 1$)
- **Cutoff energy:** 80 Ry (wavefunction), 320 Ry (charge density)
- **K-point mesh:** $8 \times 8 \times 1$ Monkhorst–Pack
- **SCF convergence:** 10^{-10} Ry (energy)
- **Force convergence:** 10^{-4} Ry/Bohr
- **2D isolation:** `assume_isolated = '2D'` (Coulomb truncation)
- **Vacuum:** 20 Å (monolayer), 25 Å (bilayer)

7.1 Computation Time

Table 8: Wall-clock times for key calculations (10 MPI processes)

Calculation	Time
Monolayer vc-relax	79 s
Monolayer SCF	~30 s
Monolayer NSCF (30 bands)	~40 s
Monolayer bands (Γ -M-K- Γ , 121 k-points)	~60 s
Strain series (7 points)	~5 min total

8 Equations and Theory

For reference, we reproduce the key equations from the original paper that govern the optical and excitonic properties (not computed in this replication).

8.1 Radiative Lifetime

The radiative lifetime of exciton state S with center-of-mass momentum $\mathbf{Q} = 0$ is:

$$\langle \tau_S \rangle = \left(\frac{8\pi e^2 E_S(0)}{\hbar^2 c} \cdot \frac{\mu_S^2}{A_{\text{uc}}} \right)^{-1} \cdot \frac{3}{4} \left(\frac{E_S(0)^2}{2M_S c^2} \right)^{-1} k_B T \quad (1)$$

where M_S is the exciton effective mass, A_{uc} is the unit cell area, and μ_S^2 is the BSE transition dipole moment.

8.2 Effective Radiative Lifetime

The thermally averaged effective radiative lifetime is:

$$\langle\tau_{\text{eff}}\rangle^{-1} = \frac{\sum_S \langle\tau_S\rangle^{-1} e^{-E_S(0)/k_B T}}{\sum_S e^{-E_S(0)/k_B T}} \quad (2)$$

9 Conclusions

We have successfully replicated the DFT-level electronic structure calculations from Bayerl et al. (2017) for H-passivated monolayer 2D GaN. Our key findings:

1. The DFT band gap of monolayer 2D GaN is **2.99 eV** (PBE), in excellent agreement with the original **2.95 eV** (LDA).
2. The relaxed lattice constant is **3.219 Å** (PBE), consistent with the expected PBE overestimate relative to LDA (3.165 Å) and experiment (3.190 Å).
3. The band gap has a direct character at Γ with near-degenerate N $2p$ valence bands, confirming the band structure topology.
4. The biaxial strain dependence of the band gap shows the same parabolic trend with an asymmetric compressive/tensile response.
5. Full replication of GW/BSE results would require BerkeleyGW and significantly more computational resources.

Replication verdict: **DFT-level results successfully replicated.** The structural and electronic properties are quantitatively reproduced within the expected differences between LDA and PBE functionals.

References

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